

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Effluents & Emission Requirements & Compliance Plan

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Rev No.	Date	Description of Revision
R01	09.08.22	Issued for Review
R02	01.11.22	Re-issued for review after incorporating TCE comments

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-HSE-PLN-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
EPA	Environmental Protection Association
HC	Hydrocarbon
VOC	Volatile Organic Compounds
FLB	Field Logistics Base
P & ID	Piping & Instrumentation Diagrams

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

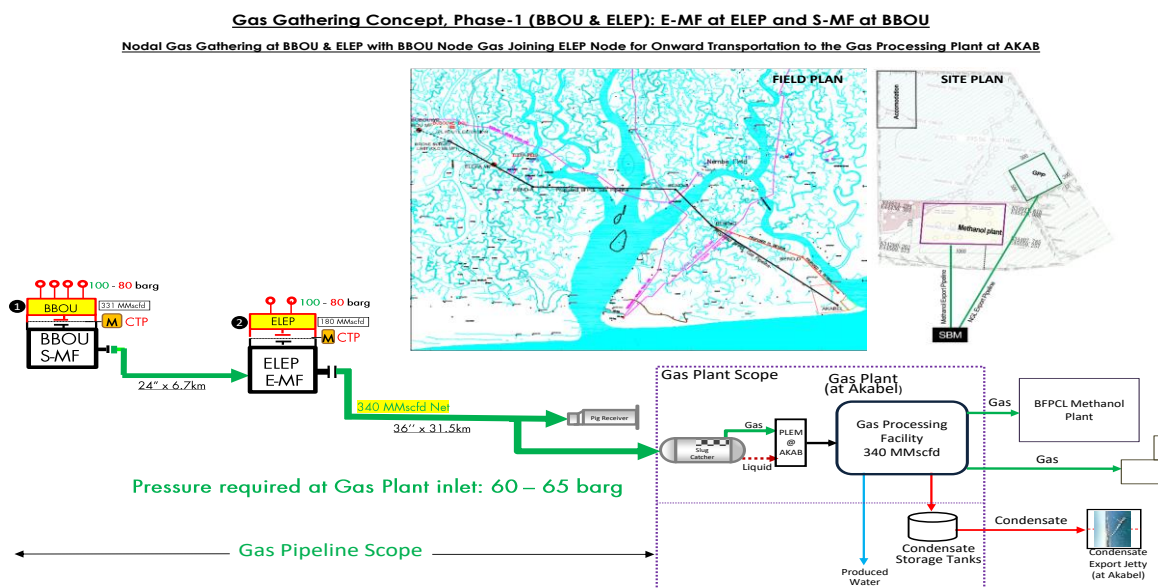


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The objective of this document is to provide an overview of the use of chemicals, emissions to air, discharges to water bodies and waste production together with the mitigation and control measures of the project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
EPA - 453/R-95-017, US EPA, November 1995	Protocol for Equipment Leak Emission Estimates

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Emissions & Discharges

3.1. Emissions

Emissions of noxious substances e.g. CO, CO₂, NO_x, SO_x, NH₃, HC are expected to occur during normal operations and maintenance activities associated with project facilities during operations. The sources of these emissions are primarily from vents at the manifolds

The main sources of noxious emissions from the project facilities will be:

- Cold vent at BBOU
- Cold vent at ELEP
- Power Generators

For additional fugitive emissions see Section 4.

To determine the impact of the plant emissions on the ambient air quality, all previous dispersion assessments on similar pipeline/manifold projects confirmed that the emissions are marginal and meet the required air quality standards.

3.2. Effluents

The notional Drainage Philosophy identifies the effluents resulting from project facilities activities and these consist of:

- Process Effluents (from pressurised or de-pressurised sources)
- Continuously Condensate Contaminated Effluents (effluents that are always considered to be contaminated - with condensate, fuel, chemicals or lubricants)
- Accidentally Condensate Contaminated Effluents (effluents that are considered to be contaminated only infrequently and by exception)
- Sewage

The systems for managing and processing effluents at the pipeline manifold stations have been described below, based on information extracted from the notional Drainage Philosophy

3.2.1. Process Effluents

At the manifolds, the pressurised process fluids (two phase fluids from blow down valves, from pressure safety valves, etc.) will be routed to the cold vent through the closed drain tanks.

The depressurised process fluids which comprise hydrocarbon liquids that may release vapour (equipment maintenance drains, pig traps, etc.) are routed to a closed drain vessel, where they are pumped into an environmental barge for discharge in the base facility. All vapour evolved

will be routed to the cold vent.

Rainfall, wash-water, condensation or other liquids which fall in the process area (potentially oil contaminated liquids that comprise mostly water) will be collected by an open drains system and pumped to the environmental barge.

3.2.2. Continuously Oil Contaminated Effluents

Continuously Oil Contaminated effluents (from drip trays, under pig-traps, chemical storage tanks) will be collected by open drains and pumped to the environmental barge

3.2.3. Oily Free Water

Oil free water from land drains outside process, roof drains etc. will be drained to the storm water drains and discharged aboard.

3.3. Solid Wastes

All solid wastes generated during operations at the manifolds will be transferred to the adjacent Field Logistics Base for re-cycling and/or disposal.

3.4. Special Wastes

Corrosion inhibition injection chemical shall be used for corrosion mitigation at the manifolds. The chemical shall be stored in storage tanks and the chemical storage shed will be roofed and bunded and chemical pumps will have drip trays, covered to prevent rain ingress. Their drain lines will normally be isolated and spillages shall be contained. An operator will periodically inspect the bund and recover any spillage or drain any rainwater.

4. Fugitive Emissions

For the project facilities, no significant amount of VOC emissions related to loading and unloading operations are foreseen, so only the fugitive emissions are estimated.

The method as described in this section is used to provide an estimate for the fugitive emissions created by the project facilities at the pipeline manifolds

4.1. Method Selection

Fugitive emissions can be estimated/measured via several methods. The United States Environmental Protection Agency (EPA) issued a Protocol for Equipment Leak Emission Estimates. According to this protocol, different approaches can be used to estimate fugitive emissions. Since no accurate data is available, the Average emission factor approach has been chosen for this project. This approach is commonly used to calculate emissions when site-specific screening data are not available, e.g. in the case of new green field developments.

4.2. Basic Assumptions

The emissions are calculated by using average component specific emission factors. The following data for a unit is required:

- a. Number and type of components, e.g. valve, pump seals, ...
- b. Medium characteristics, e.g. gas, light/heavy liquid, ...
- c. Time period, e.g. the time the component is in service

The basis for the average emission factors is the EPA report [1]. In Table 2.4 of the EPA report, the average emission factors are listed for Oil and Gas Production Operations. The factors are for the total organic compound emission rates and apply to offshore facilities, light and heavy crude facilities, gas plants and gas production facilities.

The equipment count used in these calculations has been derived from Piping and Instrumentation Diagrams (P & IDs). Based on the P & IDs, equipment likely to cause fugitive emissions e.g. valves, flanges, have been identified. For the calculations the weight percent of the medium for all equipment is taken as 100 and it has been assumed that all equipment will constantly be in operation (8760 h/year), which will most likely not be the case.

The process drains and utilities connections contain double block and bleed connections combined with spectacle blinds or spades, they are not open to atmosphere and therefore not considered in these calculations.

4.3. Results

A summary of the total fugitive emissions calculations for the project facilities is given in Table 4.1 below

Table 4-1: Fugitive Emission Calculations

Equipment	Service	No of Sources	Emission Factor (kg/h/source)	Total Emission (kg/h)
Valves	Gas/condensate	22	0.0045	0.099
Flanges	Gas/condensate	36	0.00039	0.01404
Total				0.11304
Annual Total (Tonnes)				1

The bases for the calculation are as follows.

- (1) **No of sources:** As explained in the preceding Section 4.2 - Basic Assumptions, the equipment count used in these calculations has been derived from Piping and Instrumentation Diagrams (P & IDs). Based on the P & IDs, equipment likely to cause fugitive emissions e.g. valves and flanges have been identified.
- (2) **Emission factor:** The basis for the average emission factors is the United States Environmental Protection Agency (EPA) report; EPA - 453/R-95-017, November 1995 - **Protocol for Equipment Leak Emission Estimates**. In Table 2.4 of the EPA report (page 2-15), the average emission factors are listed for Oil and Gas Production Operations. The factors are for the total organic compound emission rates and apply to offshore facilities, light and heavy crude facilities, gas plants and gas production facilities.

5. Measures To Reduce Emissions, Discharges and Wastes

The mitigation and control measures for reducing or eliminating emissions, discharges and wastes are detailed below.

5.1. Mitigation and Control Measures for Gaseous Emissions

Impairment of air quality from the project facilities, including occasional non-routine gas venting will be mitigated and controlled by the following measures:

- a. Process equipment is designed to prevent/minimize unnecessary hydrocarbon emissions.
- b. Plans will be in place to ensure regular maintenance services/checks of plant performance are undertaken by operators and a plan developed for monitoring emissions of fugitive gases from valves and pipe joints.
- c. Ambient air quality shall be monitored in line with existing regulatory requirements
- d. Vent sizing will be optimised to meet noise criteria at the edge of the sterile area(s) and the plant boundary fence.
- e. Relief valves are normally closed and only open in an emergency. They are of a type that will reseal during after operating to minimise leakage.
- f. Blow down valves are normally closed in service and will only be utilised to depressurize sections of plant to the cold event in an emergency.

5.2. Mitigation and Control Measures for Effluent Discharges

The identified impacts from effluent discharge during operations will be mitigated and controlled by the implementation of the following measures:

- a. Sanitary wastes and sewage from the manifolds shall be collected in the environmental barge and treated in the facilities at the Field Logistics Base (FLB).
- b. Emergency response spill control/prevention equipment shall be stored ready for use at the FLB.
- c. All discharge point sources shall be registered with DPR and shall ensure that effluents from construction sites are monitored and treated to comply with regulatory limits before disposal
- d. A closed drain system is provided to ensure all hydrocarbon contaminated effluent are collected and pumped into an environmental barge for disposal at the FLB.
- e. Open drains effluent containing mainly storm water will be collected and pumped into the environmental barge for disposal at the FLB

5.3. Mitigation and Control Measures for Solid Waste

The identified impacts from generation of solid wastes during operations will be mitigated and controlled by the implementation of the following measures:

- a. Generated solid waste should be segregated collected and sent to the FLB for proper segregation and disposal according to FLB waste management guidelines.
- b. A standard solid waste collection and segregation area shall be provided on site.
- c. Safe Handling of Chemical "SHOC" cards shall be provided on all sites where chemicals are handled e.g. corrosion inhibition chemical.
- d. A site specific solid waste management plan in line with FLB waste management guidelines shall be put in place before operations commence.